

The ATM Forum's private network/network interface

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The ATM Forum completed the first version of its private network/network interface (PNNI) in March 1996 and many ATM switch manufacturers are now offering early implementations. PNNI offers a different type of internetwork or internodal interface from the traditional Signalling System No 7 based approach favoured to date by 'public' network operators, such as BT. In spite of its name, however, PNNI may find its place in network service provider networks as well as in 'private' or customer networks. Some of the perceived limitations of the current specification for such an environment are currently being addressed in version 2, tentatively due for completion in the second half of 1998.

The PNNI really consists of two parts, a signalling protocol based around the ATM Forum's UNI signalling specification, and a dynamic source routeing protocol. This paper aims to provide an overview of the functionality and mechanics of PNNI, and to compare and contrast with the functionality offered by ITU-T's B-ISUP.

1. Introduction

To date, a large proportion of asynchronous transfer mode (ATM) connections, particularly in the wide area environment, have been of a permanent nature — permanent virtual circuits (PVCs) — requiring management intervention for set-up and tear-down. However, switched virtual circuits (SVCs) offer the ability to set up and tear down connections of a range of desired characteristics on demand, to a reachable end user. The private network/network interface (PNNI) [1] is just one of the signalling protocols that has been defined to do this. Others are described elsewhere in this publication.

PNNI consists of two protocols, namely a signalling protocol and a routeing protocol. The signalling protocol is based on the ATM Forum user/network interface (UNI) signalling, which in turn is related to ITU-T Q.2931 [2]. Modifications have been made to provide a network/network interface (NNI), rather than UNI, to make use of the routeing protocol, and to provide certain other extra functionality. The PNNI routeing protocol uses similar concepts to the open shortest path first (OSPF) protocol used within IP networks. It allows QoS-based dynamic hierarchical source routeing and provides for a hierarchical and scalable network.

2. PNNI routeing

In contrast to the 'traditional' approach within telecommunications networks of using static routes, PNNI operates a dynamic source-based routeing protocol. As the name implies, source-based routeing allows the originating

or source node to select the complete hierarchical path from itself to the destination, and all the intermediate nodes have to obey the source node routeing instructions. In order to achieve reliable source-based routeing, topology information related to address reachability and current resource capabilities of the nodes has to be distributed around the network on a regular basis. As observed above, PNNI is intended to be a very scalable protocol and this scalability is achieved primarily via hierarchy and summarisation. In theory, PNNI can support over a hundred levels of hierarchy, although it is difficult to envisage a scenario where more than a handful of levels would be necessary or desirable.

2.1 PNNI topology construction — the lowest level

A PNNI network is arranged into a series of peer groups (PGs). The highest level of the hierarchy will be represented by a single PG, while at the lowest level of the hierarchy, there will be many PGs. The members of the lowest level PG may correspond to actual ATM nodes, whereas the higher level PG members are 'logical' nodes or abstractions of the lower PGs. This notion of PGs enables PNNI to be more scalable.

Neighbouring nodes within a PG exchange different packet types in order to distribute topology information reliably. These packet types are:

- hello packets,
- PNNI topology state packets (PTSPs),

- database summary packets,
- PNNI topology state element (PTSE) request packets,
- PTSE acknowledgement packets.

'Hello' packets are exchanged between neighbouring nodes on a particular predefined VC — VCI = 18. 'Hello' packets are exchanged on every virtual path (VP), contained within a physical link. These 'hello' packets serve a number of purposes. Firstly, they allow the nodes to discover the identity of their neighbours, to determine whether those neighbours are actually members of the same PG and to verify the status of the physical links between the nodes. Nodes which have links to other nodes in other PGs are known as border nodes. These border nodes are responsible for creating uplinks, which provide connectivity between PGs at higher levels of the hierarchy.

Once the physical link is established and operational, topology information regarding the current capabilities of the nodes and links, along with reachable destination addresses from within the PG, can be exchanged. This exchange of information is crucial if PNNI is to perform reliable source-based routing. This information is flooded through the PG in PNNI topology state elements encapsulated in PNNI topology state packets.

Topology information includes details about both nodal and link state parameters. These parameters include values of the bandwidth and quality of service (QoS) available across links and nodes, as well as policy-based parameters. Topology information exchanges continue as long as the PNNI systems operate. It should be pointed out that PTSEs are only exchanged within PGs and not between different PGs. Only 'hello' packets are exchanged between different PGs (see Fig 1).

Once a node has learnt of the existence of other neighbouring nodes within its PG, it starts a database exchange process to synchronise databases. The exchange process involves a master node sending database summary

packets to a slave node. The packets contain summarised details of the PTSEs stored in a node's topology database. When one of the nodes receives a database summary packet, it checks its database against those PTSEs listed in the database summary packet. If the database does not contain a PTSE or if it is out of date, the node requests that its neighbour sends the relevant PTSE(s), making use of the PTSE request and PTSE acknowledgement packets.

Each PG elects a peer group leader (PGL). At start-up, each node is configured with a variable, known as the PG leader priority, which relates to the node's 'likelihood of becoming PGL'. On the basis of this variable, the nodes within a PG elect the node with the highest PG leader priority value to become the PGL. It is the responsibility of the PGL to represent its PG further up the hierarchy.

A new PTSE will be generated to update the node's topology database whenever a 'significant' change occurs (for example, as a result of call set-ups/clear-downs, topology changes, or network failures). The nature of a 'significant' change is suggested within the PNNI specification.

Additionally, nodes check the age of the PTSEs stored in their database. This ageing function decrements the PTSE 'remaining lifetime' to reflect how long they have been stored in the database. The ageing function also detects when the self-originating PTSEs are to be refreshed and when non-self-originated (i.e. received) PTSEs are to be deleted from the database as they have reached their expiry time. The expired PTSEs cannot be used for determining routing.

2.2 Building the hierarchy past the first level

The peer group leader (PGL) represents the PG as a logical group node (LGN) at the next level of hierarchy (i.e. its parent PG). The LGN represents a summarised view, or abstraction, of reachability and topology information pertaining to the child PG. It is important to remember that

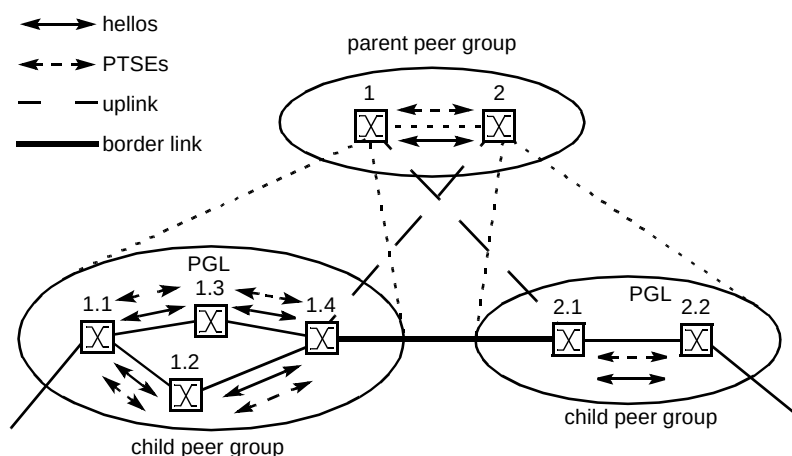


Fig 1 Exchange of 'hello' packets in the PNNI hierarchy.

the LGN is not an extra physical node, it is only an abstraction of the child PG. The LGNs use the same process of exchanging information between themselves and their peers as happened at the lower level. The LGNs in the parent PG access their peers via the lower level border nodes creating uplinks to them. The nodes in the parent PG will establish SVCs between themselves for routing information exchange. This SVC is known as an RCC and is used to exchange summarised addresses and aggregated topology information between the nodes in the parent PG. Once the databases are synchronised, PTSEs are flooded back down the hierarchy to the child PGL, which then distributes them among the PGs. This ensures that each node in the child PG has topology information not only about its own PG but also about the parent PG too. In this way, each node now has reachability information on how to access every node encompassed by the parent PG (see Fig 2).

A grandparent PG may be constructed 'above' a collection of parent PGs, and a great-grandparent, and so on, to extend the network to multiple levels of hierarchy.

2.3 Reachability and address summarisation

In addition to topology state information, address reachability information is also exchanged. In order to reduce the amount of information that needs to be distributed around the network, addresses which are reachable from a particular node are summarised. This summarisation corresponds to a 'common' address prefix. There may be a number of summarised addresses within a node, and additional addresses which share no commonality with any other address on the node. These are known as foreign addresses and are distributed intact. At the next

hierarchical layer addresses are summarised further where possible and the information distributed around that PG. This continues right up to the highest PG. The summarised addresses are flooded down the hierarchy as before so that each node has summary information about where to route calls with which address prefixes.

PNNI is defined to make use of the ATM end-system address (AESA) scheme described in UNI 3.1 [3]. These addresses are based on the ISO NSAP (network service access point) scheme and are 20 octets in length. There are three NSAP schemes described in UNI 3.1 — the International Code Designator scheme, the Data Country Code scheme, and the NSAP E.164 scheme (in contrast to 'native' E.164). All three consist of a 13-byte prefix, a 6-byte end-system identifier (ESI), and a 1-byte selector. In PNNI the 13-byte prefix ideally reflects the network topology. The ESI consists of some 'flat' address space to identify the destination UNI (a MAC address would be an example). PNNI routing is not based on the selector byte.

Each PG has an identity (the PGID) which corresponds to a prefix on an ATM NSAP type address of up to 13 bytes. The hierarchy can therefore be up to 105 levels deep. Each node also has a node ID and an ATM address corresponding to the physical system implementing the node (since nodes above the lowest level of hierarchy are abstractions) which is used to set up the SVCs for exchange of routing information in PGs above the lowest level.

Clearly, an appropriate address structure within the network is an aid to conciseness and brevity of updates. While completely unrelated addresses can be supported at any node, they must be distributed individually and

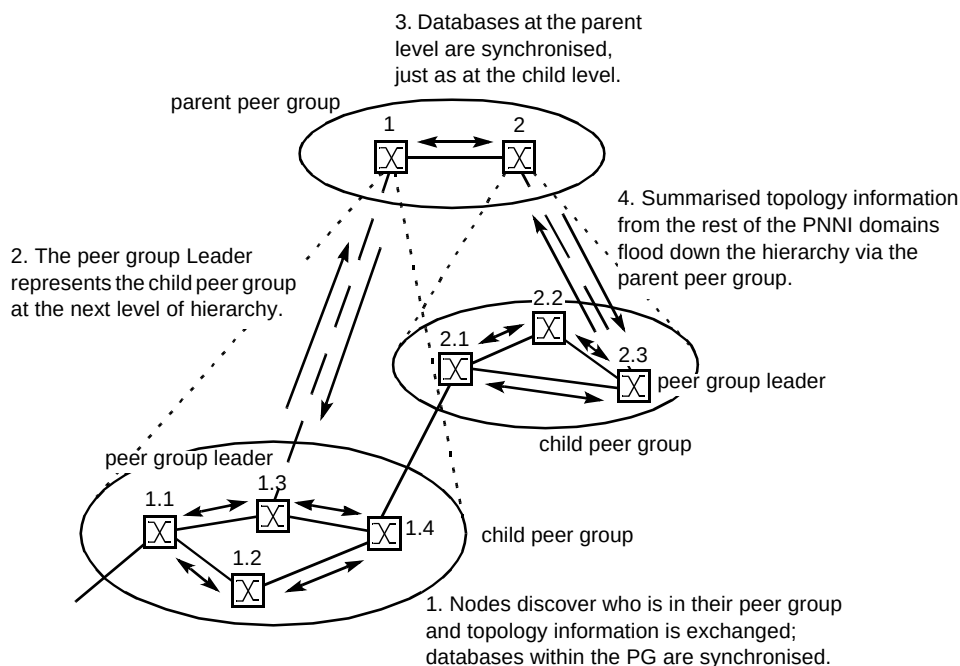


Fig 2 PNNI routing hierarchy construction in a 2-level hierarchy.

unsummarised potentially right up, down, and across the network. Thus if a chaotic address scheme is adopted, route table space is quickly filled and updates are all the more lengthy.

More information on how to address the entities in a PNNI network is given in the PNNI specification, annex F [1].

2.4 Route computation

As ATM connections can vary enormously in their requirements, such as traffic types, bandwidth, and quality of service, calculating a hierarchical source-based route (in contrast to performing hop-by-hop routing) to the destination which matches the connection requirements is a complex activity. Route computation in PNNI can be based on a number of criteria, namely QoS, reachability and a generic call admission control (GCAC) algorithm, introduced in PNNI, and described later in this section.

PNNI selects a path to a destination which has the longest address prefix match. In a large PNNI network there can be a large number of possible combinations of routes to the destination. It is the responsibility of the routing algorithm to find an acceptable route through the network. To do this the routing protocol needs to know the characteristics of the required connection, plus the capabilities of the nodes in the network. The connection characteristics can be obtained from the IEs within the set-up message, whereas the network resources are obtained from the PNNI topology information.

There are two types of route calculation that can be performed within PNNI. The first is to use an algorithm which calculates all possible destinations from a node, based upon varying connection characteristics. This is known as a pre-calculated route. Pre-calculated routes are designed to reduce some of the processing overhead involved in calculating acceptable routes. The second type are on-demand routes; these are used when the required set-up values cannot be met by any of the pre-calculated routes.

One of the factors ATM manufacturers use to differentiate their products is call admission control (CAC). Thus PNNI cannot predict what the admission criteria will be at every node. Additionally, further down the designated transit list (DTL), the nodes do not correspond to actual physical switches, but rather are logical representations based on summarised information for which a completely accurate CAC algorithm, which guarantees that the call will be admitted, cannot be run. PNNI therefore introduces the concept of generic CAC. Generic CAC is executed on receipt of a request to set up a connection in order to find which nodes are likely to admit the connection. As GCAC only predicts that a node is 'likely' to accept the call, it is

not completely accurate and the call may need to be 'cranked back' at some stage along the route.

The possible list of paths to the destination is first reduced to the links that are likely to accept the new connection, using GCAC. Path selection is via a routing algorithm and utilises required performance constraints (such as timing or loss criteria) configured within the network. Switches within a network can use different route-selection algorithms.

Once the route has been calculated it can then be converted into a DTL. These DTLs are explained later. Like CAC, route computation is not standardised and can be used by switch vendors to differentiate their products.

3. PNNI signalling

PNNI signalling is designed to be compatible with UNI 3.1 but also provides functionality from UNI 4.0 signalling, including:

- point-to-point and point-to-multipoint calls,
- parameterised quality of service,
- anycast,
- signalling for ABR,
- carriage of generic identifier transport,
- switched virtual path connections,
- frame discard,
- negotiation of traffic parameters — peak/sustained cell rate (PCR/SCR), maximum burst size (MBS),
- support for 64-kbit/s circuit mode services.

Additionally, PNNI builds on the capabilities of UNI 4.0 signalling to provide soft permanent virtual circuits (S-PVC), at both VP and VC level. Also due to the source-based routing of PNNI, the signalling capabilities support the use of designated transit lists and crank-back procedures. These additional capabilities are described below.

3.1 Call set-up and source routing

The DTL is formulated by processing the PNNI routing topology information. A DTL is basically an array of node IDs, with a pointer which is used to indicate the current node ID. The DTL IE provides a route across the PNNI domain from the ingress node to egress node, and is only carried within the SETUP message. The DTL is arranged hierarchically to provide an exact hop-by-hop route across the local PG and a hierarchical list of LGNs at each higher level within the PNNI hierarchy, which must be traversed in order to reach the destination.

The DTL procedures are best explained using Fig 3 as an example together with the stages below.

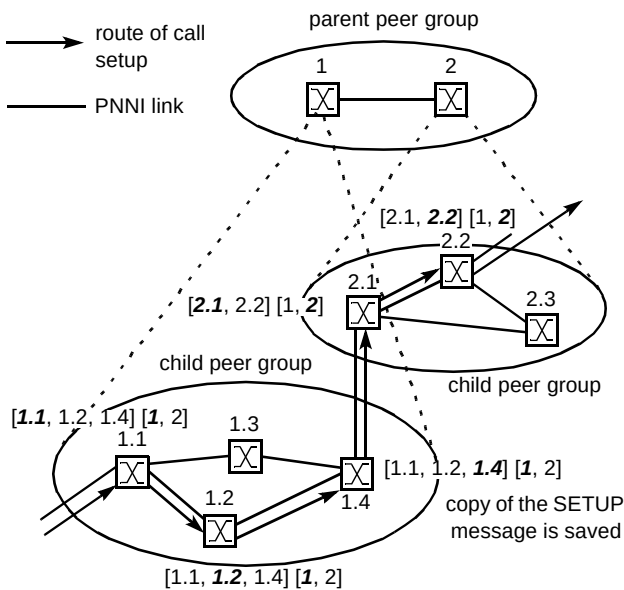


Fig 3 Call set-up in a PNNI network.

- Terminal A sends a SETUP containing called party number using UNI signalling, towards node 1.1.
- Node 1.1 calculates the DTL from itself to the destination, based upon the called party number. Node 1.1 knows the complete path across its PG, which is included in the DTL as 1.1, 1.2 and 1.4. The node 1.3 was not chosen as from its topology information it could not meet the desired characteristics of the connection. The path across the rest of the network has to be recorded using the next level PG, i.e. node 1, then node 2.
- Once the SETUP message is received at node 1.4, the first level of hierarchy within the DTL is exhausted. The pointer is incremented and points to the next level of hierarchy, namely PG 2. From the 'hello' packets exchanged between border nodes (i.e. 1.4 and 2.1), node 1.4 knows how to route to node 2.1. A copy of the entire set-up message, including the old DTL, is saved at egress to the previous PG in case of crankback.

Each time a call set-up moves from one PG to another, the DTL is recalculated using the specific information available in the new PG on how to cross it. Therefore node 2.1 knows how to traverse its PG, and from the called party number can calculate the path to terminal B, using node 2.2.

3.2 Crankback

Crankback is a facility which allows, at connection establishment time, the network to re-route a call around a failed or congested node(s), instead of clearing the call back

to the calling party. When a call cannot be processed according to the DTL, it is cleared back to the creator of that DTL (i.e. a previous node) with an indication of the problem. This indication is carried in the crankback IE within the RELEASE or RELEASE COMPLETE message. Thus the switch resources are released and no tromboning of the call occurs.

The node which created the DTL now has the responsibility of finding an alternative route around the failed node/link and to the destination. The alternative route must support the original requested connection characteristics. If this node cannot find an alternative route, the call is cranked back again towards the source node, to the next previous node which created a DTL and this node tries to route the call.

This process of crankback and re-routing continues until the source node is reached. If no alternative path can be found, the call is released.

The types of causes for crankback include resource unavailability and network faults.

3.3 Soft permanent virtual circuits

PNNI allows for the set-up of permanent virtual circuits (PVCs) via the control plane. Such connections are known as soft permanent virtual circuits (S-PVCs) and may be of VC or VP nature. An S-PVC is very similar to a PNNI switched VC (SVC) in that it is source routed and makes use of PNNI routing to find an acceptable route in real time (see Fig 4). However, the two major differences are that S-PVCs terminate within the network, thus do not cross an UNI, and are initiated by management rather than the user.

Soft permanent virtual circuits should look no different, from the customer's point of view, to the classical management-controlled PVC common in ATM networks today. Soft PVCs can be configured to automatically re-route on network failure, and offer a standards-based way to provide connectivity across a multivendor domain.

Two extra information elements have been defined to support soft PVCs. These are the calling party soft PVPC or PVCC information element and the called party soft PVPC or PVCC information element. These information elements allow the set-up message to identify the VPI/VCI combinations (or in the case of an S-PVPC, the VPI only) at the originating and terminating end-points of the soft PVC. Where the terminating end-point is not specified, the allocated VPI/VCI is returned in the CONNECT message.

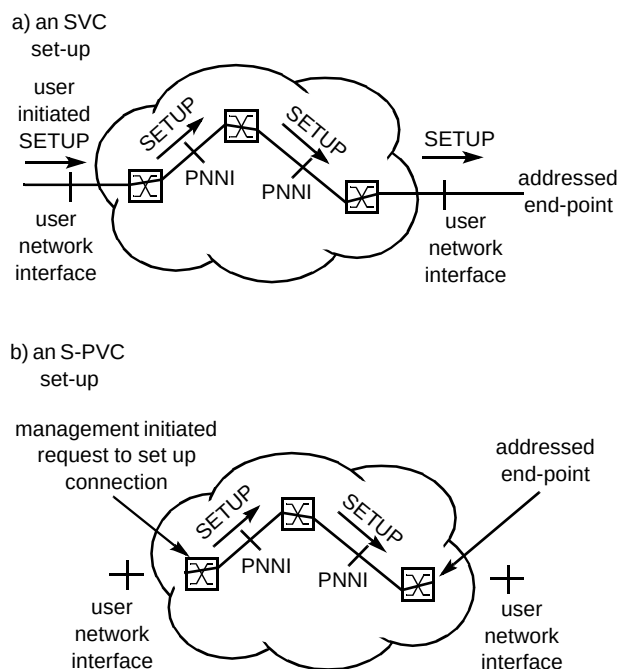


Fig 4 SVC versus S-PVC.

4. Using PNNI in the public network

Notwithstanding its name, the private network/network interface, can be used in networks by public network operators (referred to in the ATM Forum as ATM service providers) as well as in 'enterprise' networks. There are a number of particular considerations in this case.

4.1 Topology information distribution

Many public network operators are unlikely to be enthusiastic about interconnecting directly with other operators using 'full' dynamically routed PNNI. This would imply an interlinked PNNI domain with associated agreements about PG levels and topology, and also implies topology state updates coming in and out of an operator's network domain. This scenario may have serious security implications, not just because it implies a certain visibility of an operator's network state, but because it could leave a network open to showers of routing update packets should an adjoining operator configure their network incorrectly or suffer a major outage. To overcome this problem PNNI does allow for domains to be interconnected in a more traditional way, via exterior routes. Exterior routes are ones over which the PNNI routing protocol is 'turned off', so that routing is static. Introduction of static routes into what is basically a dynamic routing domain requires care to prevent introducing routing loops, since dynamic topology changes elsewhere in the network will not be tracked by the manually configured static routes.

Where dynamic routing information is exchanged, it is possible to restrict the scope of advertisements so that

certain reachability information is not distributed throughout the entire PNNI domain.

4.2 Restricting PNNI information

To help limit the advertisement of information, PNNI has a number of capabilities designed to restrict the use of certain information. There are routing constraints which can be used to restrict the selection of certain paths, links or nodes to be used. In addition, summary addresses can be suppressed, to stop the advertisement of addresses with prefixes. Suppressed summary addresses can be applied to both interior and exterior addresses. Finally PNNI supports point-to-multipoint (p-mp) connections, and a branching flag can be used to stop particular nodes from acting as a replication/branching point within p-mp calls.

4.3 Support for supplementary services

The ITU-T have defined a number of supplementary services (Q.2730/Q.2951.x) for use within a public network. These include number presentation and addressing services such as calling line identification presentation (CLIP), direct dialling in (DDI), sub-addressing (SUB) and multiple subscriber number (MSN), security services, such as closed user group (CUG), and others. Many of these are supported in UNI 4.0. However, work to support some of these in PNNI is still in progress.

4.4 Address support

PNNI is defined to support AESAs based on the ISO NSAP scheme. This is in contrast to the public telephony numbering scheme based on Recommendation E.164 formulated by the ITU-T. Whether they choose to use them to address their customers or not, the majority of public network operators are likely to want to support 'native' E.164-style addresses, if only for interconnect and interoperability. Some equipment vendors may choose to implement PNNI in such a way as to support native E.164 addresses directly — they could also be supported by mapping in and out of NSAP format as appropriate.

4.5 Support for intelligent network services

Intelligent network (IN) type services (such as number translation) are typical of the sort of facility provided by a public network. However, private networks typically do not use the classic ITU-style IN. Provision of IN-like services is possible with PNNI but is outside the scope of this paper. Network intelligence is examined again in section 5.7.

4.6 Interworking with ITU-T/access signalling

PNNI signalling, like UNI 4.0 signalling, is based around ITU-T's broadband access signalling system, Q.2931 [2], and refers back to it on a paragraph-by-

paragraph basis for much of its functionality. This is not to say, however, that there are not some notable differences, many of which are covered in section 5 below. This means that interworking issues need to be borne in mind during implementation.

4.7 *Call correlation across multiple networks*

In order to correlate information across different nodes within a network (for billing and reporting), or across multiple networks (e.g. for reciprocal billing arrangements) the network call correlation identifier may be used. This has only recently been defined in the standards bodies, and is not explicitly supported in PNNI version 1, although it could be transported using the generic identifier transport (GIT) information element, which is supported in PNNI version 1. Generation and transport of unique NCCIs, particularly across multiple network domains, has formed the basis of discussions during formulation of PNNI version 2.

4.8 *Closed user groups*

Many large corporate business customers require enhanced security features over and above a basic any-to-any switched ATM service. The requirement might include a need to screen callers from outside their business from accessing their equipment and end-stations, or to prevent calls being placed outside the company. In its most basic form, such security can be provided by screening calling and called party numbers at connection set-up time, either within the public network, or on the customer premises equipment. This requires a matrix of calling and called party numbers to be held at each node, either for allowed calls, or for barred calls. This may be onerous to administer.

Closed user group defines a procedure which simplifies this process. Instead of having to check at each call set-up whether a specific address is allowed to call another specific address, each user is assigned membership of a 'closed user group', with a particular unique ID of its own. At call set-up a check is only made as to whether that caller is a member of the correct CUG, so that at network ingress information about far-end addresses need not be kept. Because the relationship between user and CUG is a many-to-one rather than a many-to-many connection, CUG is easier to administer and is more scalable than straight address-screening tables. However, the signalling capability required to provide closed user groups is only presently being defined, partly because of the original thrust of PNNI as a private network interface.

In the future, security features like closed user groups could be provided on ATM networks using intelligent network functionality.

4.9 *Scalability*

Scalability is clearly of paramount importance in a public network. As described above, PNNI has been designed with the aim of being a very scalable dynamic routing protocol. In order to achieve this scalability, information about the network is aggregated. Of necessity, however, this aggregation must lead to some reduction in accuracy of information in proportion to the degree of abstraction. PNNI is theoretically able to support a large number of peer group levels, likely to be well in excess of that required for a world-wide public network. However, practical limits imposed on aggregation are likely to be much lower.

5. **Comparison with ITU-T network signalling**

The most traditional choice for a public network operator is often to comply with international recommendations as published by the ITU-T. The ITU-T has defined the broadband ISDN user part (B-ISUP — Q.2761-4 inclusive [4]) for layer 3 network signalling. In the first instance, this is based very much on its narrowband equivalent, ISUP, (Q.761-4 inclusive). For more details regarding B-ISUP, see Jones [5].

5.1 *Protocol Stacks*

As can be seen from Fig 5, both stacks make use of the same lower layer ATM protocols. The service-specific control function (SSCF), service-specific connection-oriented protocol (SSCOP) and ATM adaptation layers (AAL) are usually combined in a single component known as the signalling AAL (SAAL). The purpose of the signalling AAL is to adapt the signalling messages into ATM cells and vice versa. It is at the higher layers of the protocol stack, where the two differ significantly. Some of these differences are explained below.

5.2 *Routing philosophy*

The routing philosophies of the two protocols are very different. This is mainly due to the types of networks in which they were designed to operate (i.e. PNNI in private networks and B-ISUP in public networks). PNNI consists of both a routing protocol and a signalling protocol, whereas B-ISUP is just a signalling protocol. In Signalling System No 7 (SS7), there is the concept of a signalling network. The maintenance of the signalling network and the routing of signalling messages is undertaken by the message transfer part level 3 (MTP3). Call routing is the responsibility of B-ISUP, and CAC is performed by exchange applications, and they are not the subject of standardisation as ITU-T leave the CAC and call-routing algorithms to be determined by the network operator/switch vendor. While it is difficult to compare the call-routing

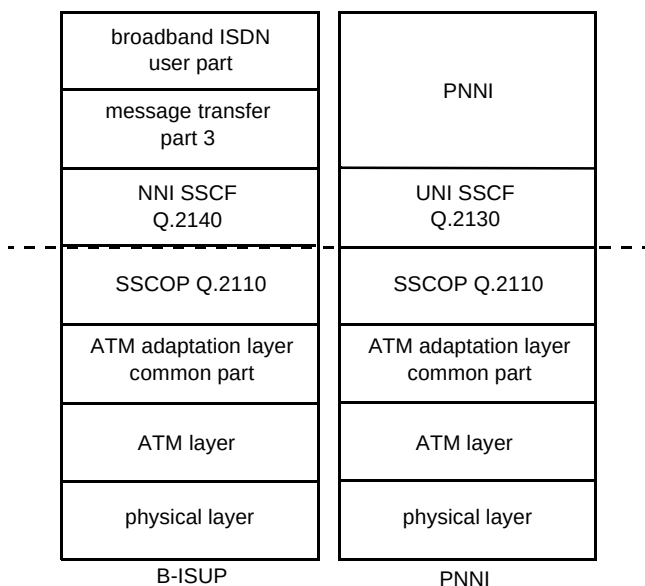


Fig 5 Comparison of protocol stacks for PNNI and B-ISUP.

methods of PNNI and B-ISUP in detail, it is worthwhile discussing the routing of signalling in both protocols.

Unlike PNNI which uses hierarchical source-based routing, MTP-3b uses hop-by-hop routing. In hop-by-hop routing, rather than the source determining the path towards the destination, it is the responsibility of each node along the path to route the signalling messages based on its own routing knowledge. As the name implies, each node only knows how to route signalling messages to the next hop along the path towards the destination, rather than the complete route. With a hop-by-hop routing algorithm there is no need to distribute topology information around the network. Instead the routing information is configured into a routing database by management. The routing information normally consists of primary routes to be used in normal conditions and secondary routes to be used in failure conditions. Primary routes are normally the shortest path to the destination (i.e. least number of hops). Therefore B-ISUP and MTP-3b have no concept of 'hello', PTSE or database summary packets, and there is no database synchronisation or PGL election processes.

In either broadband or narrowband SS7 networks, two different types of signalling modes can be used, to route signalling messages between nodes. These two modes are known as associated and quasi-associated modes of signalling. Quasi-associated signalling is a form of non-assigning signalling. PNNI only supports the associated mode of signalling. Associated signalling is the more direct form of signalling between nodes. In this mode, the signalling messages are sent over a signalling link, which directly interconnects the signalling points. Quasi-associated signalling makes use of signalling transfer points (STPs). One or more STPs are used to interconnect the signalling points (see Jones [5] for diagrams of these signalling modes).

Whereas PNNI can use dynamic routing, for the routing of signalling messages, MTP-3b employs a more static or predetermined routing approach. Each node determines the outgoing link over which the signalling message will be sent based upon a destination point code (DPC) and optionally a predetermined signalling link selector (SLS) using a routing table stored either within each node or remotely. With this static routing method, the decision to select a signalling link is not affected by the signalling traffic loading, unless signalling congestion or failure occurs on those outgoing links.

Recent studies [6] have indicated that routing algorithms should have a close relationship with the node's CAC algorithm. This enables the routing algorithm to predict the probability of a node accepting the connection. In PNNI, this is provided by the use of the GCAC in-route computation, as described previously in section 2.4. SS7 networks do not have this concept of GCAC, where information related to the status of the node's probability of connection acceptance is flooded around the network.

One of the challenges facing network operators intending to implement source-based routing is to try to reduce the time delay in computing on-demand routes, while at the same time managing the amount of topology-related information being distributed between nodes within the network and preventing routes being selected without the awareness of the operator.

5.3 Behaviour under abnormal conditions

Due to the different backgrounds, PNNI and B-ISUP differ in the level of resiliency they provide. As B-ISUP and MTP-3b were designed for use within a public network they need to be robust to cope with failures and congestion within a network. One of the major parts of the MTP-3b functionality is to provide signalling network management. This tries to ensure the integrity of the signalling network under failure or abnormal conditions. The signalling network management is split into three sub-parts, known as signalling traffic management, signalling route management and signalling link management. These are responsible for diverting signalling traffic from failed routes to operational routes, without message loss, managing signalling congestion, and determining the availability and status of signalling routes, along with controlling signalling links. Therefore if a signalling link fails after the call/connection has been established, provided there is another available signalling link, the signalling messages can be diverted and control of the call maintained.

Using B-ISUP and MTP functionality, the amount of signalling traffic transmitted within the control plane can be managed. If a node receives more signalling messages than it can process, an indication is sent to the signalling user part causing this overload, instructing it to back off until the

congestion is cleared. This type of control plane congestion is not presently available in PNNI.

PNNI was originally developed to be used within private networks, and as such takes a different approach to availability and resilience. PNNIv1 has a crankback functionality, which can be used only at connection set-up to route around congestion and/or failure along the selected route. As mentioned previously, PNNI supports soft PVCs to provide resiliency for PVCs in the case of failure. PNNI v2 should contain procedures to provide fault-tolerant routing. This would allow a failed connection to be automatically re-routed from the source node using a different DTL, without any intervention from the end users.

5.4 Quality of service

The ITU-T has defined a series of QoS classes which are sets of values of delay and loss, and which are held in the QoS parameters IE. In contrast, the ATM Forum has gone one step further, in allowing individual specification of these parameters (seven in total) — all held in the extended QoS parameters IE. Unlike B-ISUP, which will only support QoS classes, PNNI supports QoS classes and optionally extended QoS parameters. The extended QoS parameters IE was one of the capabilities carried forward from ATM-F UNI v4.0. They consist of both desired and cumulative cell delay variation (CDV) and an acceptable cell loss ratio (CLR) in the forward and backward directions. The cumulative QoS values allow each switch involved in a connection to add its expected QoS impairments to a running total. If the calling user's requested parameter values have not been exceeded when the connection reaches the called user, the connection is established, otherwise the connection is released.

5.5 Addressing

B-ISUP currently supports native E.164 addresses and optionally NSAP E.164s. Work is under way to allow B-ISUP to support other types of NSAP formats.

5.6 Scalability

It can be argued that both protocols scale to support very large networks. In an SS7 network, nodes are identified by point codes (PCs). Therefore, theoretically the size of an SS7 network is limited by the number of PCs available. The world-wide signalling network is split into two independent levels, namely the international and national levels. The format of the PCs is also split into two levels, i.e. international and national, to reflect this. These PC values are administered on a world-wide basis by the ITU-T [7]. SS7-based networks have proved to be scalable as they are used the world over today. As PNNI is not based upon a traditional SS7 network, it does not make use of PCs. Instead, nodes within the network are identified by NSAP

addresses. With PNNI, the use of hierarchy levels, allows for the aggregation of topology information and address summarisation, which enables PNNI to support a large number of nodes. Although PNNI has not yet been deployed on a world-wide basis, with careful use of hierarchy levels for the aggregation of topology information and address summarisation, PNNI should be able to support a large number of nodes.

5.7 Network intelligence and mobility

Again, due to the history of the two protocols, their support of aspects of intelligent networks (INs) and mobility differ. For the integration of IN and broadband, B-ISUP will be used along with the services of transaction capabilities (TC) and signalling connection control part (SCCP). For the integration of mobile and broadband, it is likely that B-ISUP, plus TC and SCCP will also be used, although no decision has yet been made. TC and SCCP are mainly used to transfer non-circuit-related data between nodes in an SS7 network. Like B-ISUP, they make use of the transfer capabilities of MTP-3b. Non-circuit-related information allows the network to retrieve data from another node/database without making a connection (i.e. connectionless transfer). Examples include database access for number translation, mobile registration/location updates and terminal capability checking, such as look-ahead.

As PNNI was originally designed for use in a private network the support of IN and mobility was not seen as essential. If, however, public network operators wish to use PNNI in their networks they would seek to integrate PNNI with their existing IN and mobility capabilities. As PNNI does not specify the use of TC or SCCP functionality, a different approach needs to be found. Although PNNIv1 does not support mobility, the ATM Forum wireless ATM working group (W-ATM) may adopt PNNI to be used in their mobility specifications. The W-ATM specification is due for completion in 1999.

5.8 Interworking

As mentioned before, B-ISUP and PNNI are very different protocols. To address this differing functionality, the ATM-F set up a technical committee known as 'Interworking among ATM Networks' (IAN) at the beginning of 1997. The motivation for this activity is that it is seen as essential for the growth of ATM to have interworking between all networks, be they PNNI or B-ISUP. The distinction between public and private networks is somewhat blurred/converging as some public networks are/will be using PNNI for initial deployment of ATM SVCs.

The IAN technical committee is defining the interface between a number of broadband networks. This interface is known as the ATM internetwork interface (AINI). The

scope of this work is shown in Fig 6. It seems unlikely that AINI will be used to interconnect two B-ISUP networks where the natural choice would be to use B-ISUP. There is still work to be done as there are many aspects of the two protocols that may not allow for easy interworking, such as the support of CUG interlock codes and network call correlation IDs (NCCIs).

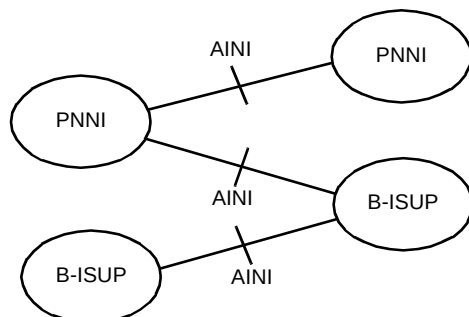


Fig 6 Scope of the ATM internetwork interface.

PNNI supports the necessary narrowband messages and IEs, allowing interworking with narrowband networks. However, there are no explicit procedures within the PNNI specification on how this is achieved.

BQSIG is the broadband equivalent of QSIG. Although QSIG was defined a number of years ago, BQSIG is still being defined by the European Computing Manufacturers Association (ECMA). Both BQSIG and QSIG are intended to be used as inter-PBX signalling within private networks and are based upon UNI signalling. ATM-F has recognised the need for this interworking and PNNIv2 will have procedures for BQSIG support.

6. Evolution

The ATM-F has developed a series of network signalling protocols, based around UNI signalling protocols. Initially they defined the interim interswitch signalling protocol (IISP), using UNI v3.1 signalling and static routing. After significant work they then released PNNI v1, which was completed in 1996. An errata version of PNNIv1 was completed in early 1997 to correct the errors. Now work is progressing on PNNIv2, which is due to be released later this year. The proposed additional capabilities of version 2 are:

- network call correlation identifier (NCCI),
- leaf-initiated join (LIJ),
- closed user group (CUG),
- authentication of routing — peer entity,
- BQSIG interworking (for private networks),
- supplementary services,
- fault-tolerant connections.

In addition, the ATM-F have been developing protocols to be used in an IP-over-ATM environment. To date the ATM Forum have started initial work on integrated PNNI (I-PNNI) and PNNI augmented routing (PAR). Both of these are based upon PNNI concepts, the idea being that PNNI routing could be used for both ATM switches and IP routers and have one integrated protocol. There are a number of issues to be solved and it remains to be seen if these ideas are adopted.

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